

Task Management REST API

A lightweight, robust, and fully documented RESTful Task Management API built with Node.js and Express.

This project provides complete CRUD lifecycle management for tasks, strictly structured JSON payloads, robust HTTP error handling, and interactive API documentation powered by OpenAPI 3.0 and Swagger UI.

Quickstart & One-Command Execution

Prerequisites

- **Node.js** (v18.x or higher)
- **npm** (included with Node.js)
- **Git**

Installation & Run (One-Liner)

Clone the repository, install dependencies, and launch the API server with a single command:

```
npm install && node server.js
```

Once running, the API daemon will listen at:

- **Base API:** <http://localhost:3000>
- **Interactive Swagger UI Documentation:** <http://localhost:3000/docs/>

API Endpoint Reference

HTTP Method	Route Path	Path Param	Request Body Payload	Success Code	Error Codes	Operation Summary
GET	/	None	None	200 OK	—	API system metadata banner
GET	/health	None	None	200 OK	—	System health check / heartbea

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GET	/tasks	None	None	200 OK	—	Retrieve full task inventory
POST	/tasks	None	{"title": "string"}	201 Created	400 Bad Request	Register a new task
GET	/tasks/{id}	id (integer)	None	200 OK	404 Not Found	Lookup task by primary key
PUT	/tasks/{id}	id (integer)	{"title": "string", "done": boolean}	200 OK	400 Bad Request, 404 Not Found	Update task title or state
DELETE	/tasks/{id}	id (integer)	None	204 No Content	404 Not Found	Purge task from memory

Terminal Verification (curl -i Output)

Below is an authentic raw HTTP protocol verification log captured directly from the daemon response using curl.exe -i:

Sample Query: POST /tasks (Resource Creation)

```
HTTP/1.1 201 Created
X-Powered-By: Express
Content-Type: application/json; charset=utf-8
Content-Length: 53
ETag: W/"35-Jz3aR7pP"
Date: Sat, 25 Jul 2026 01:56:00 GMT
Connection: keep-alive
```

```
{"id":4,"title":"Automate Kenyan SME invoices","done":false}
```

Sample Query: GET /tasks/4 (Record Query)

HTTP/1.1 200 OK

X-Powered-By: Express

Content-Type: application/json; charset=utf-8

Content-Length: 53

ETag: W/"35-Jz3aR7pP"

Date: Sat, 25 Jul 2026 01:56:10 GMT

Connection: keep-alive

```
{"id":4,"title":"Automate Kenyan SME invoices","done":false}
```



Interactive Swagger UI (/docs)

The API exposes an interactive developer interface rendered from an **OpenAPI 3.0 specification** (openapi.json).



Tech Stack

- **Runtime Environment:** Node.js
- **Framework:** Express.js
- **API Blueprint:** OpenAPI 3.0 Standard (openapi.json)
- **Interactive Engine:** swagger-ui-express
- **Version Control:** Git & GitHub